

MODULE 00 · COURSE-WIDE

Advanced SQL for Analytics

Thirteen hours, twenty-six modules, one fictional grocer, and a database you can break.

PART [start here](#) MODULE 00 of 26 RUNS about 12 min

A query that runs is not a query that is right. **Nothing in SQL tells you the difference.**

So the course is arranged around that

- ◆ **Every technique is taught alongside how it fails**, because the failure is the part you will meet and the syntax is not.
- ◆ **Ninety-two worked scenarios**, each one a plausible wrong number, with five seconds to diagnose it before the answer.
- ◆ **And an assertion at the end of every module**, which is the one line that would have caught it.
- ◆ **It is not a reference course**. Syntax is lookupable, and this spends its time on the part that is not.

THE TAKEAWAY

If you want a catalogue of window functions, the documentation is better than I am and it is free. This is about what happens when you use them on real data.

The eight parts

PART	WHAT IT COVERS, AND ROUGHLY HOW LONG
1 Foundations	Clause order, determinism, nulls and ties. ~1h
2 Windows	Frames, ranking, navigation, distributions. ~3h 45m
3 CTEs and recursion	Materialisation, hierarchies, cycles and cost. ~1h 45m
4 Sequencing	Gaps and islands, sessions, intervals, as-of joins. ~1h 50m
5 Reshaping	Pivot, unpivot, unnesting, semi-structured data. ~1h
6 Sets and grouping	Set operations, grouping sets, rollup and cube. ~55m
7 Proof and performance	Assertions, query plans, cost and incrementality. ~1h 40m
8 Practice	Diagnostics, twelve unsolved, the capstone, and a close. ~1h 25m

THE TAKEAWAY

Part 2 is by far the largest, because window functions are where most analytical SQL lives and where most of it goes wrong.

You write SQL already. This assumes joins, GROUP BY, and a subquery hold no fear.

What it assumes, precisely

- ◆ **Joins, aggregates, GROUP BY, subqueries.** Used comfortably, without looking them up.
- ◆ **Some exposure to a warehouse**, or at least to a table with more rows than fit on a screen.
- ◆ **No window functions required.** Module three starts from what a window actually is.
- ◆ **And no particular engine.** Examples run on DuckDB; dialect differences are called out where they matter.

THE TAKEAWAY

The course covers PostgreSQL, Redshift, Snowflake, BigQuery, SQL Server, Oracle and DuckDB, and is explicit about which of them lack which feature.

Kestrel Foods. **412 stores, 1.2 million members**, and fourteen deliberate defects.

Some of what is wrong with Kestrel's data

THE DEFECT	MEASURED
Transactions with lines tied inside one second	30.8%
Sales lines carrying no promotion	62.0%
Sales lines with no member (guest shopping)	38.0%
A cycle in the category hierarchy	42 categories unreachable
Ledger postings back-dated, median lag 22 days	14.0%
Ranged products that have never sold	exactly 1,400

THE TAKEAWAY

These are counted from the shipped database, not asserted. Every percentage quoted in this course is a number you can run a query against yourself.

The data is **generated, not described**. If I claim a query fails, you can watch it fail.

Where the scenarios come from

- ◆ **They are the shapes real defects take**, not puzzles. A report 3% below finance, two systems 77% apart, a total that is exactly 3x.
- ◆ **Almost none of them is an error message**, because a query that errors has already told you something.
- ◆ **Each has four candidate causes, and usually two are plausible**, which is the part that makes them worth pausing on.
- ◆ **And the wrong answers are explained too**, because knowing why a good theory fails is most of the skill.

THE TAKEAWAY

Every solution slide says why each of the three wrong options is wrong. Those are often more useful than the right one, and they are why the scenarios take a minute each.

How a module is built

- ◆ **The mechanism first**, worked on five or six rows you can check by hand, rather than on a million.
- ◆ **Then four scenarios**. A realistic wrong number, four candidate causes, and five seconds to commit.
- ◆ **Then the dialect differences**, because they are real and they are the part most likely to have changed.
- ◆ **And five habits at the end**, which are what actually survive.

THE TAKEAWAY

Every module ends with the same two slides: what people commonly believe that is wrong, and the five habits. Those are the parts to rewatch.

When a slide says **PAUSE AND DECIDE**,
pause. The five seconds is not enough, and
it is not meant to be.

What ships with the course

- ◆ **kestrel.duckdb**, at 170 MB, with all fourteen defects at their real proportions. One pip install and it runs on a laptop.
- ◆ **A capstone brief**, with ten deliverables, and a harness of 39 assertions that marks your answers.
- ◆ **A worked solution**, which you open after the harness passes and not before.
- ◆ **And an answer file for module twenty-four**, whose twelve diagnostics have no answers in the video.

THE TAKEAWAY

The database is the durable part. The slides will fade; a file containing fourteen real defects is something you can point a query at in two years' time.

The harness checks **properties, not values**.
It can tell you exactly what is wrong
without telling you what is right.

How to watch this

- ◆ **In order, for parts 1 to 3.** Later modules lean on them heavily and will not make sense otherwise.
- ◆ **Parts 4 to 7 are more independent**, so skip to what you need and come back.
- ◆ **Not at 2x.** The scenarios need thinking time, and speeding up a pause is a strange thing to do.
- ◆ **And in sittings.** Thirteen hours is a term's worth of evening classes, not an afternoon.

THE TAKEAWAY

Parts 1 to 3 are about 6 hours and everything after them assumes all of it. If you only do one thing, do those in order.

If you only have an hour

- ◆ **Module 02, determinism, ties and nulls.** The single highest-value half hour here, and it costs nothing to apply.
- ◆ **Module 05, the frame clause,** if you write window functions at all. Most people have one specific belief about frames that is wrong.
- ◆ **Or module 20, proving a query right,** if you would rather have the discipline than the techniques.
- ◆ **Any of those three stands alone,** which is not true of the rest.

THE TAKEAWAY

Module 02 is the recommendation. Almost every defect in the other twenty-four modules has something to do with a null, a tie, or an ordering that does not identify a row.

What this course is not

- ◆ **Not physical design.** Distribution keys, sort keys, clustering and file layout are touched and not taught.
- ◆ **Not streaming SQL,** where windows run over unbounded input and a watermark means something else entirely.
- ◆ **Not geospatial, full-text or vector search,** each of which is its own subject with its own operators.
- ◆ **And not orchestration.** Incrementality is covered; how you schedule and test a pipeline is not.

THE TAKEAWAY

Each of those is a course. Knowing where the edges are is part of knowing what you have, which is why this slide is at the front rather than buried at the end.

And what will have dated

- ◆ **Every engine support table in here.** Redshift gained three clauses in 2023 alone; Oracle gained four in 21c.
- ◆ **Features under discussion as this was recorded**, including IGNORE NULLS and row pattern recognition on PostgreSQL.
- ◆ **So check before designing around a gap**, because a missing feature is the least durable fact in this course.
- ◆ **The behaviours will not date.** Nulls, ordering, frames and grain are in the standard and have been for decades.

THE TAKEAWAY

The dialect tables are the perishable part. Where this course disagrees with your engine's documentation, the documentation is right and I am out of date.

Every number quoted in this course was
**measured against the database that ships
with it.**

Which is also how to read it

- ◆ **Confident delivery is not evidence.** I am confident for thirteen hours and I was wrong about at least one thing.
- ◆ **The measured claims are load-bearing**, and they come with a query you can run.
- ◆ **The unmeasured ones are opinions**, including most of the advice about what you should do.
- ◆ **And the database is there so you can check me**, which is a better arrangement than my authority.

THE TAKEAWAY

The distinction worth holding throughout: which claims came with a number and a reproduction, and which came with a confident tone.

Most of these defects were found by
somebody who was not looking for them.

Not twenty-six techniques. **About seven habits**, and the reflex to ask how you would know.

Three minutes of setup, before module one

```
1 pip install duckdb
2
3 # download kestrel.duckdb from the course pack, then:
4 python3
5 >>> import duckdb
6 >>> con = duckdb.connect('kestrel.duckdb')
7 >>> con.execute('SELECT COUNT(*) FROM fact_sales_line').fetchone()
8 (7069736,)
9
10 # that is the whole setup. No warehouse, no account, no cluster.
```

8 Seven million sales lines on a laptop. If this returns, you can reproduce everything in the course.

THE TAKEAWAY

One dependency and one file. A course whose examples need a platform is a course most people never run anything from, which is why DuckDB was chosen.

Module one is next, and it is about the order a query actually runs in.